

EXHIBIT 37

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U.S. Patent No. 9,865,546 Infringement Chart

Title: Contacts to semiconductor substrate and methods of forming same

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Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	Tower '546 Accused Process
[1pre] A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:	To the extent the preamble is limiting, devices manufactured by Tower that rely on its SiGe and BiCMOS processes (including the Tower SBC18 process) (hereinafter, "Tower '546 Accused Process") comprise a method for forming a contact in a dielectric layer to a semiconductor substrate.¹ For example, the Toyota Blind Spot Monitor Sensor (#88162-0R031) includes a module manufactured by DENSO Corporation (identified as DNSRR004), which further includes a die manufactured by Tower (identified as M7DN25PS) that is, on information and belief, representative of devices manufactured by Tower that rely on its SBC18 process.

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '546 Accused Processes and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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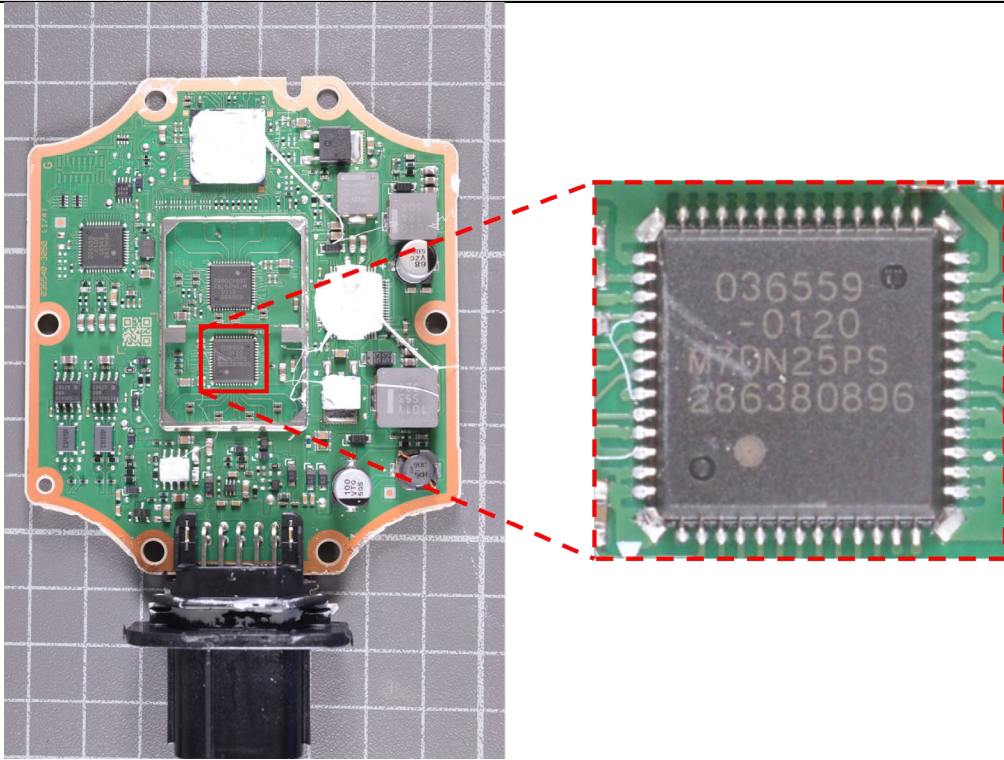
Claim	Tower '546 Accused Process
	 To the extent the preamble is limiting, the Tower '546 Accused Process comprises forming a contact in a dielectric layer to a semiconductor substrate, as explained further in the limitations below.
[1a] forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. For example, Tower '546 Accused Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. The reverse engineering analysis below shows the contact opening after it has been filled (in black).

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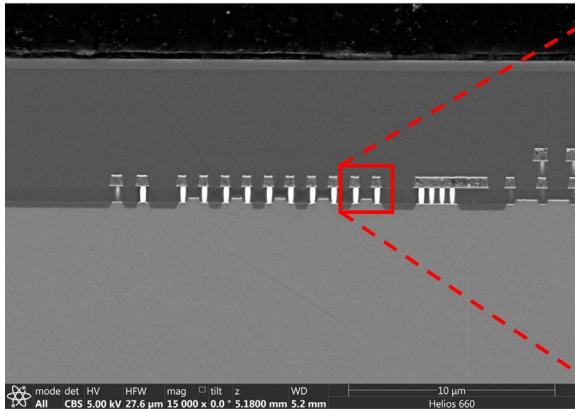
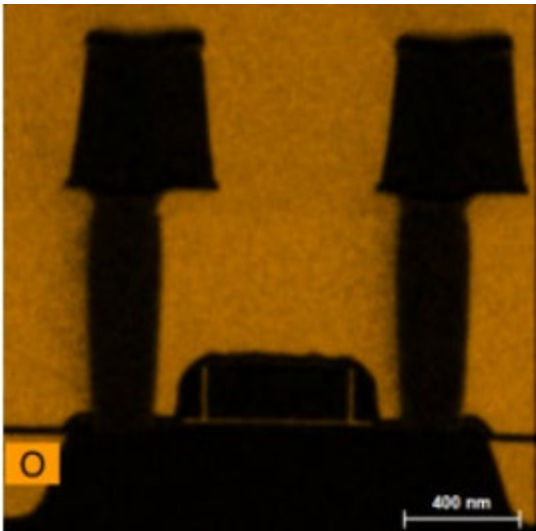
Claim	Tower '546 Accused Process
	 Scanning electron micrograph (SEM) showing a series of vertical structures on a substrate. A red box highlights a specific contact opening. The image includes technical data at the bottom: mode det HV HFW mag tilt z WD 10 µm Helios 660. All CBS 5.00 kV 27.6 µm 15 000 x 0.0 ° 5.1800 mm 5.2 mm.  Magnified SEM image of the contact opening, showing two vertical structures on a substrate. A scale bar indicates 500 nm. As shown below, the contact opening is formed in a dielectric layer.  Scanning electron micrograph (SEM) showing two vertical structures on a substrate. A scale bar indicates 400 nm.

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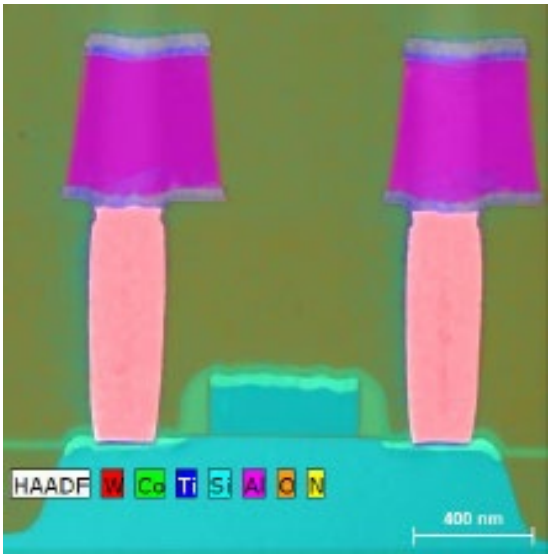
Claim	Tower '546 Accused Process
[1b] depositing a first liner layer to conformally coat the contact opening;	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat the contact opening. For example, the Tower '546 Accused Process comprises depositing a first liner layer to conformally coat the contact opening.  The image is a cross-sectional scanning electron micrograph (SEM) of a semiconductor device. It shows two vertical structures, likely contact openings, side-by-side. Each structure has a blue top layer, a red middle layer, and a green base layer. A scale bar at the bottom right indicates 400 nm. A legend at the bottom left shows the following elements: W (red), Co (green), Ti (blue), Si (purple), Al (yellow), and N (orange).

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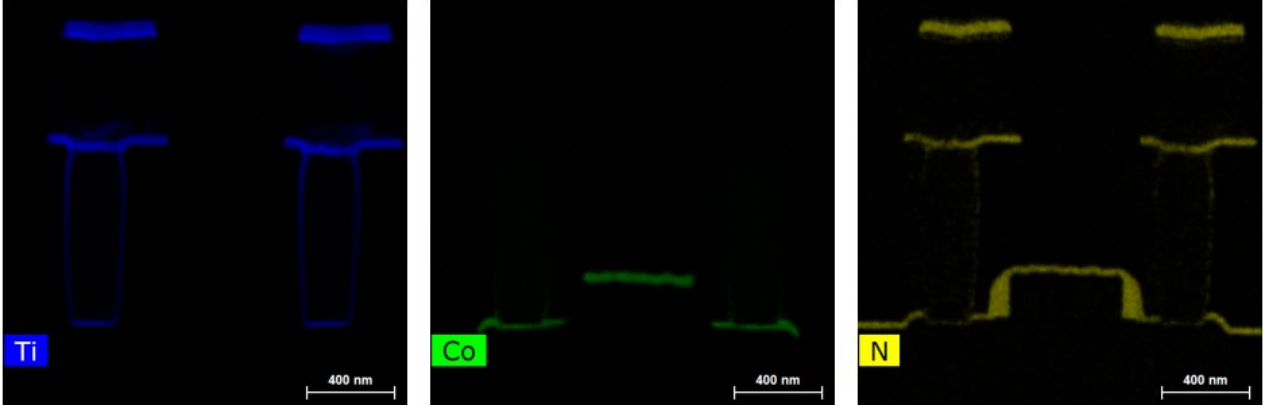
Claim	Tower '546 Accused Process
	
[1c] causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;	The Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate. For example, the Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate, as shown in the image below. The diffused portion of the first liner layer forms a first intermix region at the upper portion of the substrate. The green arrow in the image below corresponds to the x-axis in the ensuing graph. That is, the upward direction in the image below corresponds to right-to-left in the ensuing graph.

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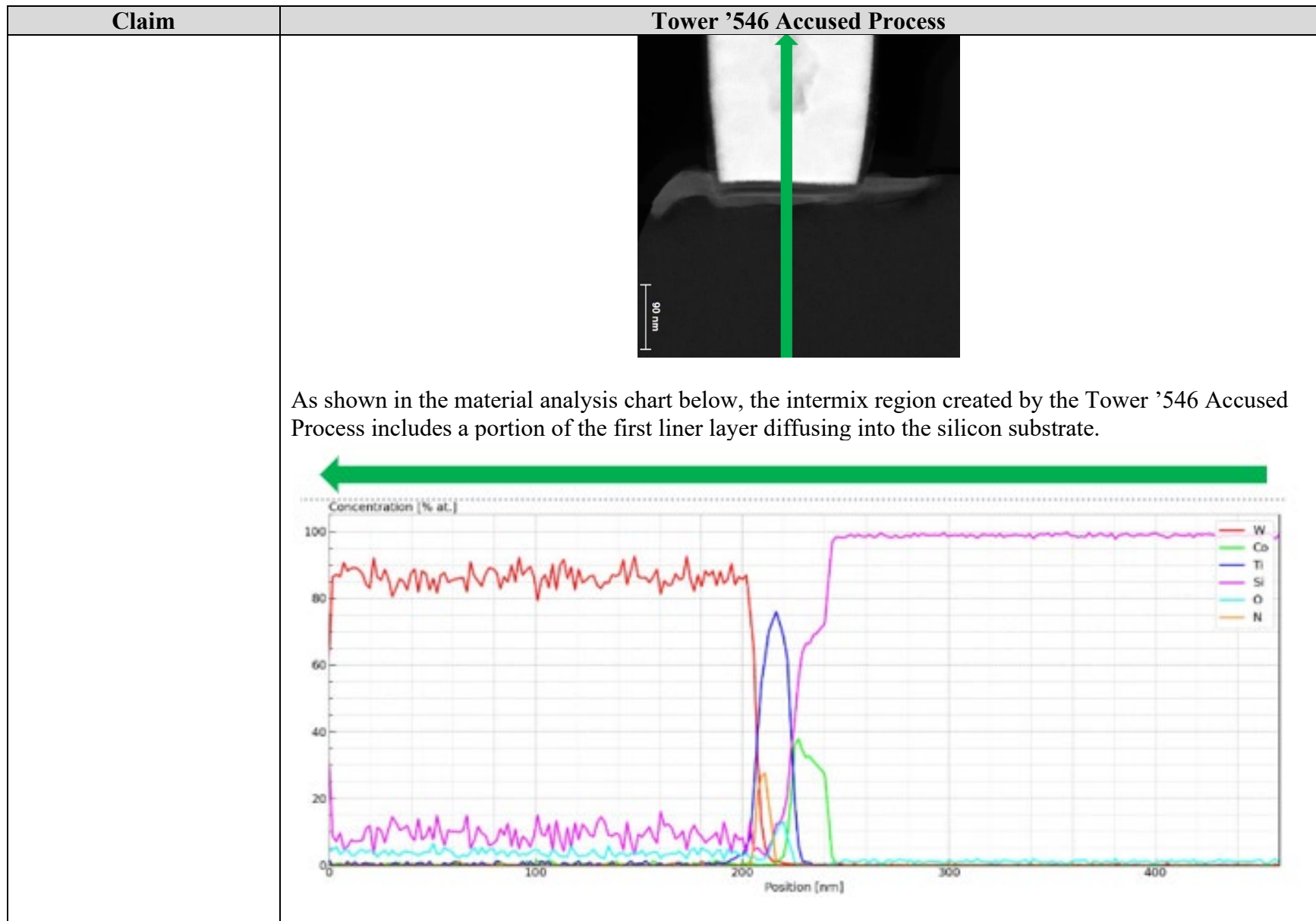


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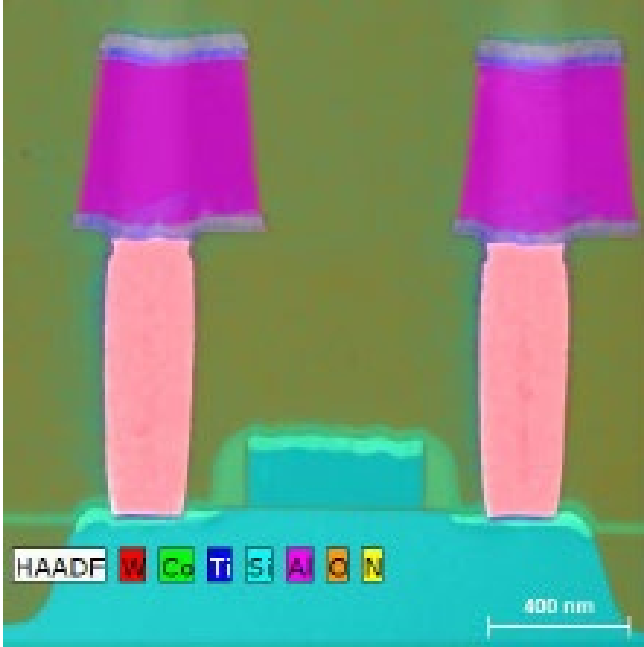
Claim	Tower '546 Accused Process
[1d] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. For example, the Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. 

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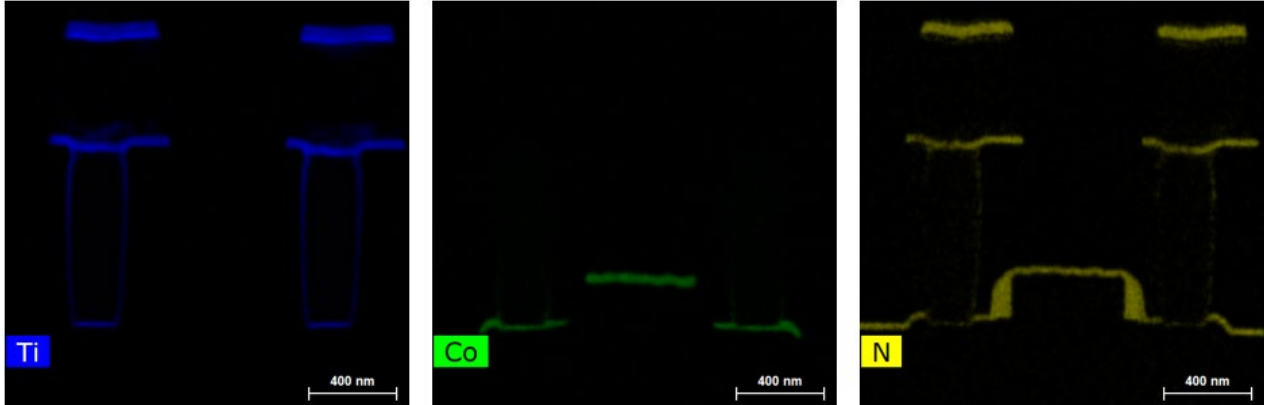
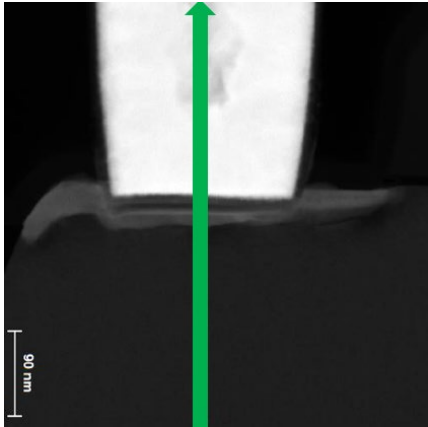
Claim	Tower '546 Accused Process
	 Again, the green arrow in the reverse engineering image below corresponds to the x-axis in the ensuing material analysis chart.  As shown in the material analysis chart below, the refractory metal layer is deposited to the left of (i.e., on top of) the intermix region.

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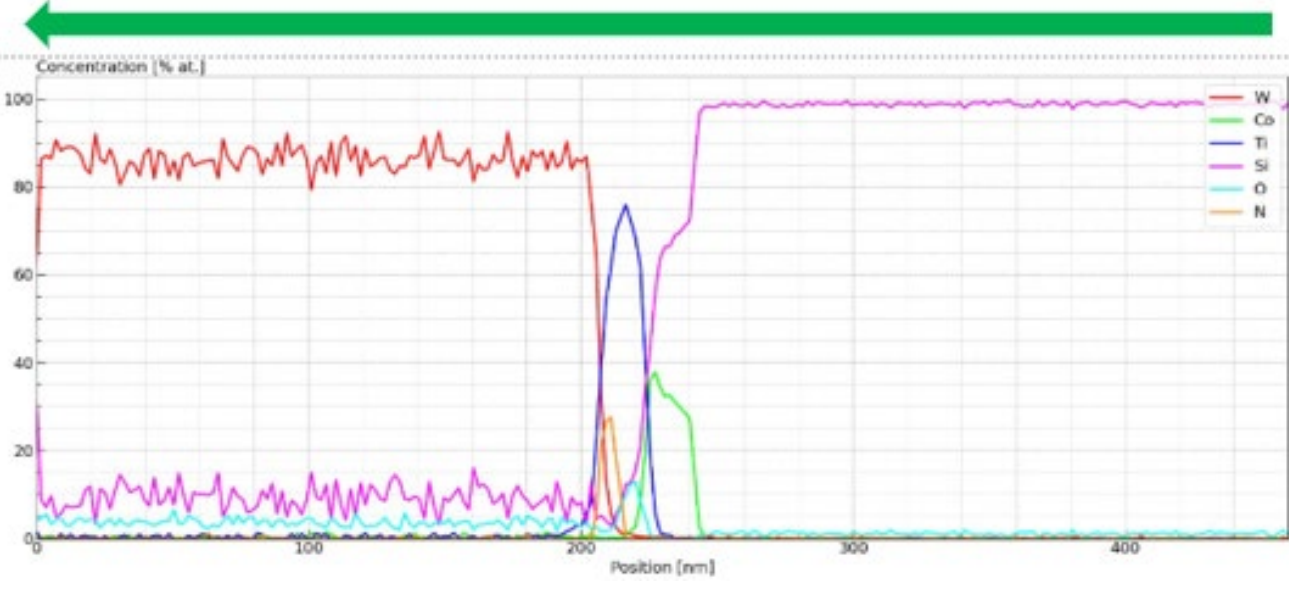
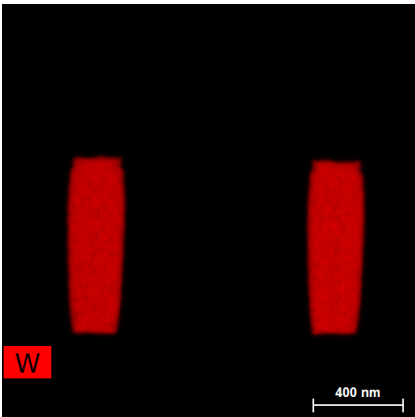
Claim	Tower '546 Accused Process
	 The graph displays the atomic concentration of various elements as a function of position. A green arrow at the top indicates the process direction from right to left. The x-axis represents position in nanometers (nm), ranging from 0 to 400. The y-axis represents concentration in atomic percent (% at.), ranging from 0 to 100. The legend identifies the elements: W (red), Co (green), Ti (blue), Si (purple), O (cyan), and N (orange). Tungsten (W) and Cobalt (Co) concentrations are high (around 80-90% at.) from 0 to approximately 200 nm, then drop sharply. Silicon (Si) and Titanium (Ti) concentrations are low until about 200 nm, where they both show sharp peaks (Si reaching ~75% at., Ti reaching ~70% at.) before dropping again. Oxygen (O) and Nitrogen (N) concentrations remain very low throughout the scan.
[1e] depositing a metal in the contact opening thereby forming the contact.	The Tower '546 Accused Process comprises depositing a metal in the contact opening thereby forming the contact.  

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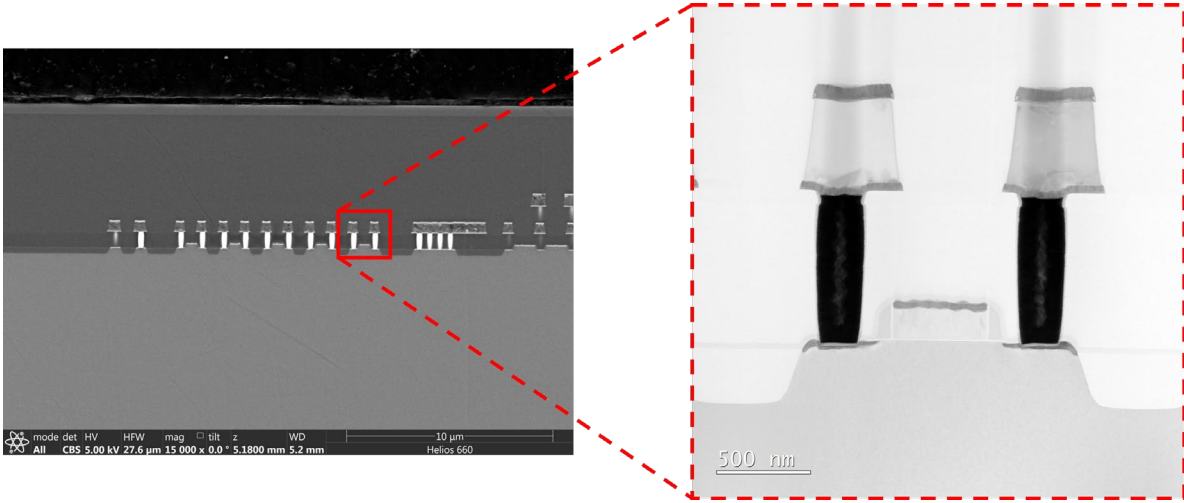
Claim	Tower '546 Accused Process
[8pre] A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:	To the extent the preamble is limiting, the Tower '546 Accused Process comprises a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. For example, the reverse engineering analysis below shows a transistor with a source (e.g., left) and drain (e.g., right) with a gate located between the source and drain. 
[8a] forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain;	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain. <i>See, e.g.,</i> limitation [1a].
[8b] depositing a first liner layer to conformally coat each contact opening and causing a portion of	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain. <i>See, e.g.,</i> limitation [1b] and limitation [1c].

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Claim	Tower '546 Accused Process
the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;	
[8c] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. <i>See, e.g.</i> , limitation [1d].
[8d] depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.	The Tower '546 Accused Process comprises depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts. <i>See, e.g.</i> , limitation [1e].